

Figure 3 : SimpleCV Blob operations

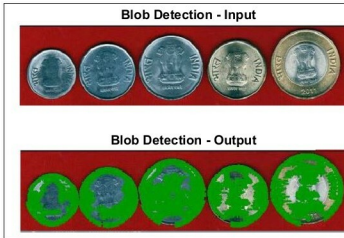


Figure 4 : Blob detection input and output images

A simple code for blob handling is as shown below:

```
from SimpleCV import Image
coins = Image("coins.png")
binCoins = coins.binarize()
blobs = binCoins.findBlobs()
blobs.show(width=8)
```

After binarizing the image, the blobs are identified by directly applying the function `".findBlobs()"`. The example program loads an image with coins in it. The `findBlobs` is used to identify the pixels representing the coins leaving the background pixels. The screenshot of the input and output images for the blob detection are as shown in Figure 4.

Motion Detection Example

Motion detection is an important task in many machine vision applications. SimpleCV allows the developer to perform the motion detection task using less than ten lines of code as shown below:

```
from SimpleCV import *
cam = Camera()
threshold = 5.0 # setting an adjustable threshold for motion detection
while True:
    previous = cam.getImage() #get the current image
    time.sleep(0.5) #pause for half a second. The 0.5 can be adjusted
    current = cam.getImage() #get another image
    diff = current - previous # compute the image difference
    matrix = diff.getNumpy()
    mean = matrix.mean()
    diff.show()
    if mean >= threshold:
        print "Some Movements Detected"
```

As it can be observed from the aforementioned code, the motion detection is carried out by subtracting the current image from the snapshot taken half a second ago. The SimpleCV facilitates direct subtraction of these image shots as shown in the code.

A detailed capability demo of SimpleCV is available at <https://youtu.be/UZSm7Q2bZoc>. In Summary, SimpleCV really makes the Vision computing easy by providing a wide collection of methods. As it is Python based, the seamless integration shall be performed with other useful Python Libraries. 

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